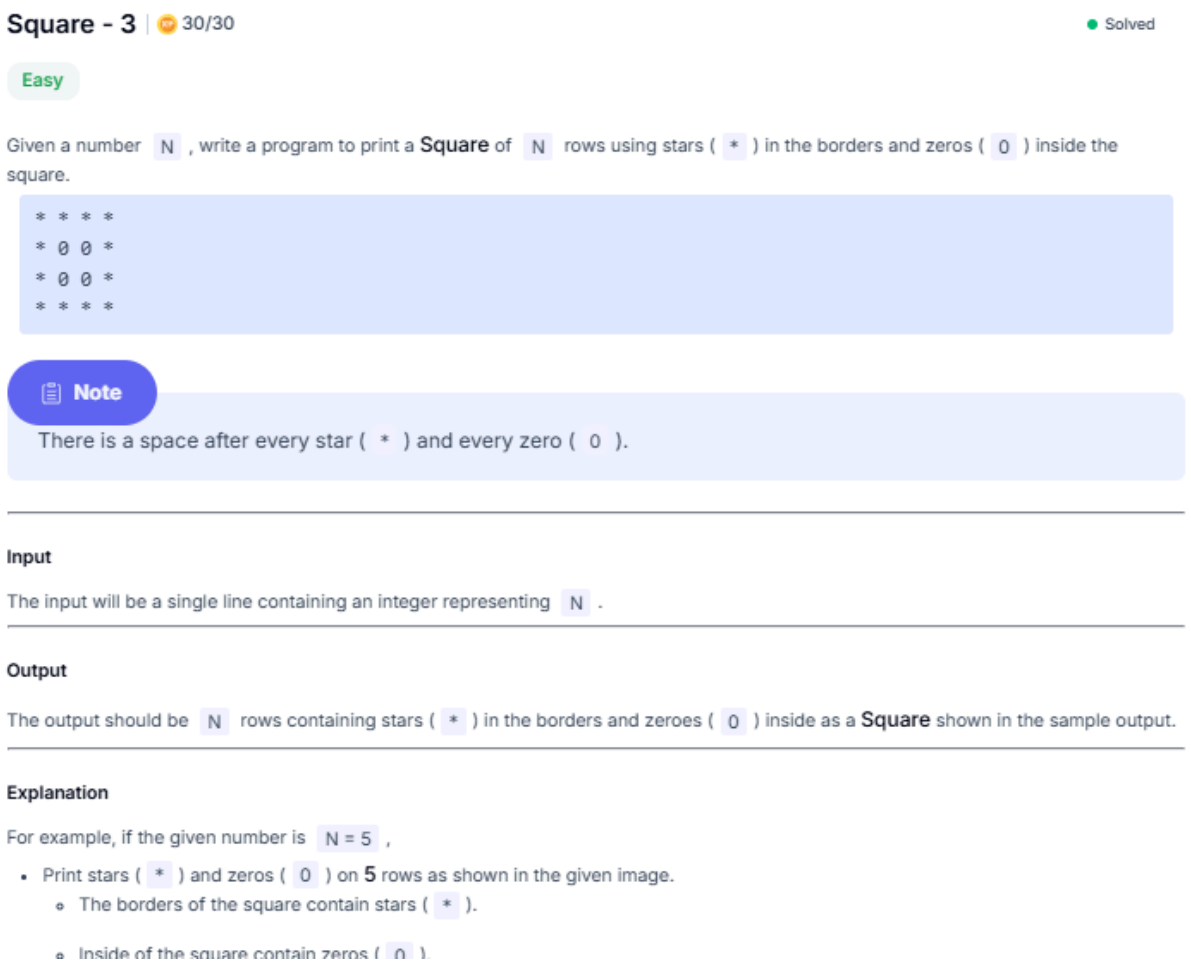
cp_10f

1



Square - 3 | 30/30 — Solved

Easy

Given a number `N` , write a program to print a **Square** of `N` rows using stars (`*`) in the borders and zeros (`0`) inside the square.

```
* * * *
* 0 0 *
* 0 0 *
* * * *
```

Note

There is a space after every star (`*`) and every zero (`0`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) in the borders and zeroes (`0`) inside as a **Square** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print stars (`*`) and zeros (`0`) on **5** rows as shown in the given image.
  - The borders of the square contain stars (`*`).
  - Inside of the square contain zeros (`0`)

code:

```
n=int(input())
for i in range(1,n+1):
    if i==1 or i==n:
        row="* "*n
        print(row)
    else:
        zero="0 "*(n-2)
```

```
        row="* "+zero+"* "
        print(row)
```

2

Rectangle - 4 | 30/30

Solved

Easy

Given two numbers `M` and `N`, write a program to print a **Rectangle** of `M` rows and `N` columns using stars (`*`) in the borders and zeros (`0`) inside the the rectangle.

```
* * * * *
* 0 0 0 *
* 0 0 0 *
* * * * *
```

Note

There is a space after every star (`*`) and every zero (`0`).

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

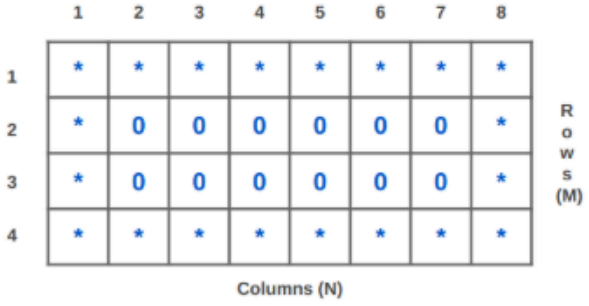
Output

The output should be `N` rows containing stars (`*`) in the borders and zeros (`0`) inside as a **Rectangle** shown in the sample output.

Explanation

For example, if the given numbers are `M = 4` and `N = 8`,

- Print stars (`*`) and zeros (`0`) on 4 rows and 8 columns as shown in the given image.
  - The borders of the rectangle contain stars (`*`).
  - Inside of the rectangle contain zeros (`0`).



Note

In the given image,

- The Zero in the box indicates zero and a space ("0 ").
- The Star in the box indicates star and a space ("* ").

code:

```
m=int(input())
n=int(input())
```

```
for i in range(1,m+1):
    if i==1 or i==m:
        row="* "*n
        print(row)
    else:
        zeros="0 "*(n-2)
        row="* "+zeros+"* "
        print(row)
```

3

Right Angled Triangle - 9 | 30/30

Solved

Easy

Given a number `N` , write a program to print a **Right Angled Triangle** of `N` rows using dots (`.`) in the borders and zeros (`0`) inside the right angled triangle.

```
.
. .
. 0 .
. 0 0 .
. . . . .
```

Note

There is a space after every dot (`.`) and every zero (`0`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing dots (`.`) in the borders and zeros (`0`) inside as a **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 7` ,

- Print dots (`.`) and zeros (`0`) on 7 rows as shown in the given image.
  - The borders of the right angled triangle contain dots (`.`).
  - Inside of the right angled triangle contain zeros (`0`).

Note

In the given image,

- The Dot in the box indicates a dot and a space (". ").
- The Zero in the box indicates a zero and a space ("0 ").

code:

```
n=int(input())
for i in range(1,n+1):
    if i==1:
        row=". "
        print(row)
    elif i==n:
        row=". "*n
        print(row)
```

```
    else:
        zeros="0 "*(i-2)
        row=". "+zeros+". "
        print(row)
```

4

Hollow Square | 35/35

Given a number `N` , write a program to print a **Square** of `N` rows using stars (`*`).

```
* * * * * *
*         *
*         *
*         *
*         *
* * * * * *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as a **Square** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print stars (`*`) on **4** rows as shown in the given image.
  - The first row contains **4** stars (`*`).
  - The middle **2** rows contain **2** stars (`*`).
  - The last row contains **4** stars (`*`).

code:

```
n=int(input())
for i in range(1,n+1):
    if i==1 or i==n:
```

```
        row="* "*n
        print(row)
    else:
        spacing="  "*(n-2)
        row="* "+spacing+"* "
        print(row)
```

5

Hollow Rectangle | 25/25

Solve

Given two numbers `M` and `N` , write a program to print a **Rectangle** of `M` rows and `N` columns using stars(`*`).

```
* * * *
*     *
*     *
*     *
*     *
* * * *
```

Note

There is a space after every star (`*`).

Input

The first line of input contains an integer representing `M` .
The second line of input contains an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as a **Rectangle** shown in the sample output.

Explanation

For example, if the given numbers are `M = 4` and `N = 6` ,

- Print stars (`*`) on **4** rows and **6** columns as shown in the given image.
  - The first row contains **6** stars (`*`).
  - The middle two (`4 - 2 = 2`) rows contain **2** stars (`*`).
  - The last row contains **6** stars (`*`).

code:

```
m=int(input())
n=int(input())
```

```
for i in range(1,m+1):
    if i==1 or i==m:
        row="* "*n
        print(row)
    else:
        spacing="  "*(n-2)
        row="* "+spacing+"* "
        print(row)
```

6

Hollow Right Angled Triangle | 30/30

Given a number `N` , write a program to print a **Right Angled Triangle** of `N` rows using stars (`*`).

```
*
* *
*   *
* * * *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as a **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print stars (`*`) on **5** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    if i==1:
        row="* "
```

```
        print(row)
    elif i==n:
        row="* "*n
        print(row)
    else:
        spacing="  "*(i-2)
        row="* "+spacing+"* "
        print(row)
```

```
7
```

Hollow Right Angled Triangle - 2 | 30/30

Solved

Medium

Given a number `N` , write a program to print a **Right Angled Triangle** of `N` rows using underscore (`_`), pipe (`|`), and forward slash (`/`).

Input

The input will be a single line containing an integer representing `N` .

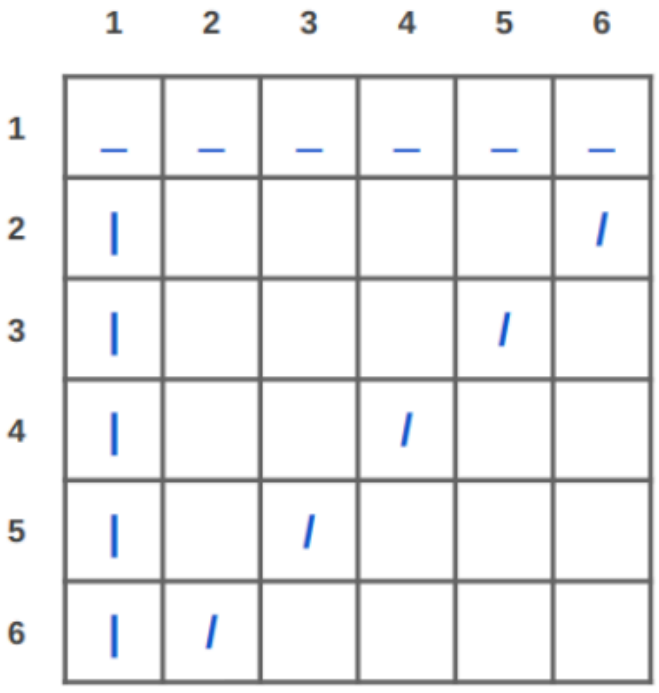
Output

The output should be `N` rows containing underscore (`_`), pipe (`|`), and forward slash (`/`) as a **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print underscore (`_`), pipes (`|`), and forward slashes (`/`) on **6** rows as shown in the given image.
  - The first row contains **6** underscores.
  - The next **5** rows contain pipe (`|`) and forward slash (`/`).



code:

```
n=int(input())
for i in range(1,n+1):
    if i==1:
        row="_"*(n+1)
        print(row)
```

```
    else:
        spacing=" "*(n-1)
        row="|"+spacing+"/"
        print(row)
        n=n-1
print("|/")
```